

Gut Microbiome Signatures of Type 2 Diabetes

Traditional Statistical Methods, and Hypothesis Testing

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Abstract

This project examines whether gut microbiome composition can distinguish patients with Type 2 Diabetes (T2D) from healthy controls. The analysis uses a cohort of 116 patients (63 T2D, 53 healthy) and combines microbiome-specific preprocessing, differential abundance testing, taxonomy-aware validation, and supervised machine learning. Raw genus- and species-level abundance features were preprocessed with MIPMLP using taxonomy level 7, sum-based aggregation, relative normalization, and rare-feature filtering, reducing the dataset to 258 taxa. We then applied a global $\log_{10}(X + 1e-6)$ transformation for visualization and downstream analysis. Differential abundance analysis with the Mann-Whitney U test and Benjamini-Hochberg FDR correction identified 16 significant taxa. A global taxonomy-aware validation with mi-Mic was not significant ($p = 0.99998$), suggesting that the T2D signal is not driven by broad tree-wide shifts. In predictive modeling, the strongest full-feature baseline was Random Forest with ROC-AUC 0.780 , followed by XGBoost with ROC-AUC 0.720 . A taxonomy-aware CNN built from image-like microbiome matrices reached ROC-AUC 0.608 , and a reduced-feature follow-up improved XGBoost to ROC-AUC 0.741 without surpassing the full-feature Random Forest result. Overall, the results support a moderate but biologically specific microbiome signature associated with T2D.

Introduction

Type 2 Diabetes is a complex metabolic disease associated with chronic inflammation, insulin resistance, and systemic metabolic imbalance. In recent years, the gut microbiome has emerged as a potential contributor to metabolic disease, raising the question of whether microbiome composition can help distinguish patients with T2D from healthy individuals.

Our project asks two related questions. First, can gut microbiome profiles separate T2D patients from healthy controls? Second, is the relevant signal best understood as a broad taxonomic shift across the microbiome or as a narrower multivariate pattern concentrated in specific taxa? These questions are clinically and biologically relevant because they address both prediction and interpretation: whether microbiome data contain useful diagnostic signal, and what kind of microbial changes underlie that signal.

The analysis follows the course framework for medical machine learning by combining preprocessing tailored to microbiome data, non-parametric statistical testing, careful evaluation with held-out test data, and comparison across multiple modeling approaches. Rather than assuming that more complex models must perform better, the project compares classical tabular models with a

taxonomy-aware CNN baseline and interprets the results in light of statistical findings.

Methods

Dataset

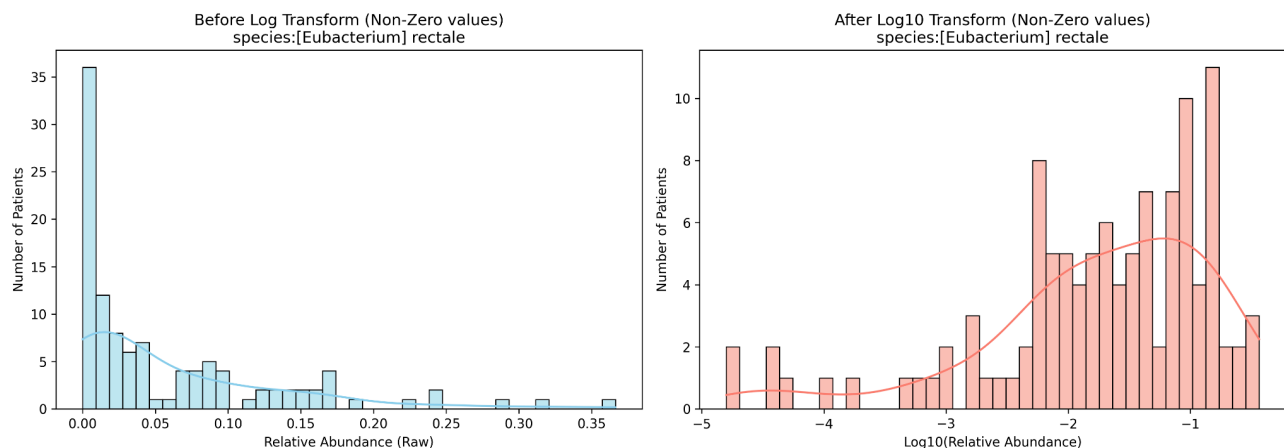
The main analysis uses ``T2D_microbiome_dataset_2.csv``. After selecting genus- and species-level microbiome columns, the working dataset contains 116 samples and 498 raw microbial abundance features. The outcome label is binary: ``T2D`` or ``healthy``.

Preprocessing

Because microbiome data are sparse, high-dimensional, and strongly skewed, we applied microbiome-specific preprocessing with MIPMLP. The pipeline used:

- taxonomy level ``7``
- taxonomic grouping by ``sum``
- relative normalization
- rare-bacteria filtering threshold ``0.05``

This reduced the feature space from 498 raw taxa to 258 processed taxa. For exploratory analysis and downstream modeling, we also applied a global ``log10(X + 1e-6)`` transformation to stabilize the scale and compress extreme abundance values.



Exploratory Analysis

We examined:

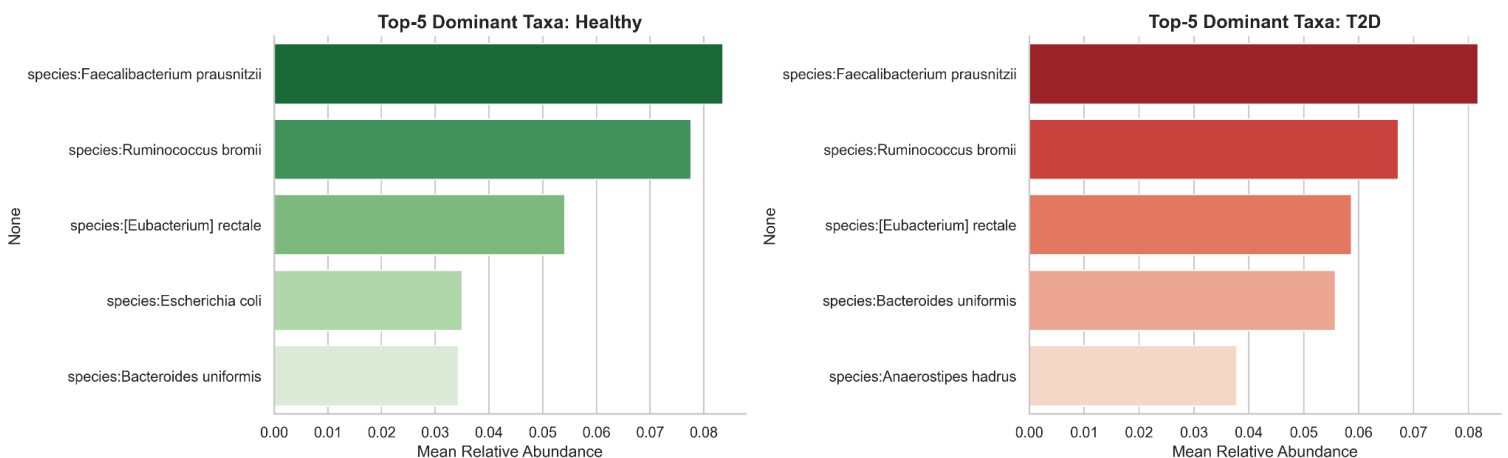
- the top-5 dominant taxa in each class based on mean relative abundance
- sparsity by counting non-zero taxa per patient
- the effect of the log transformation on highly variable taxa

These steps were used to characterize the microbiome structure before formal testing and modeling.

Statistical Analysis

To identify taxa associated with T2D, we performed a Mann-Whitney U test independently for each processed taxon, comparing the healthy and T2D groups. Because many taxa were tested in parallel, p-values were adjusted with Benjamini-Hochberg FDR correction. For interpretation, we also computed log2 fold-change values to quantify the direction and magnitude of each association.

To test whether the signal reflected a broader hierarchical microbiome shift, we applied the taxonomy-aware `mi-Mic` procedure and used its nested ANOVA output as a global structural validation step.



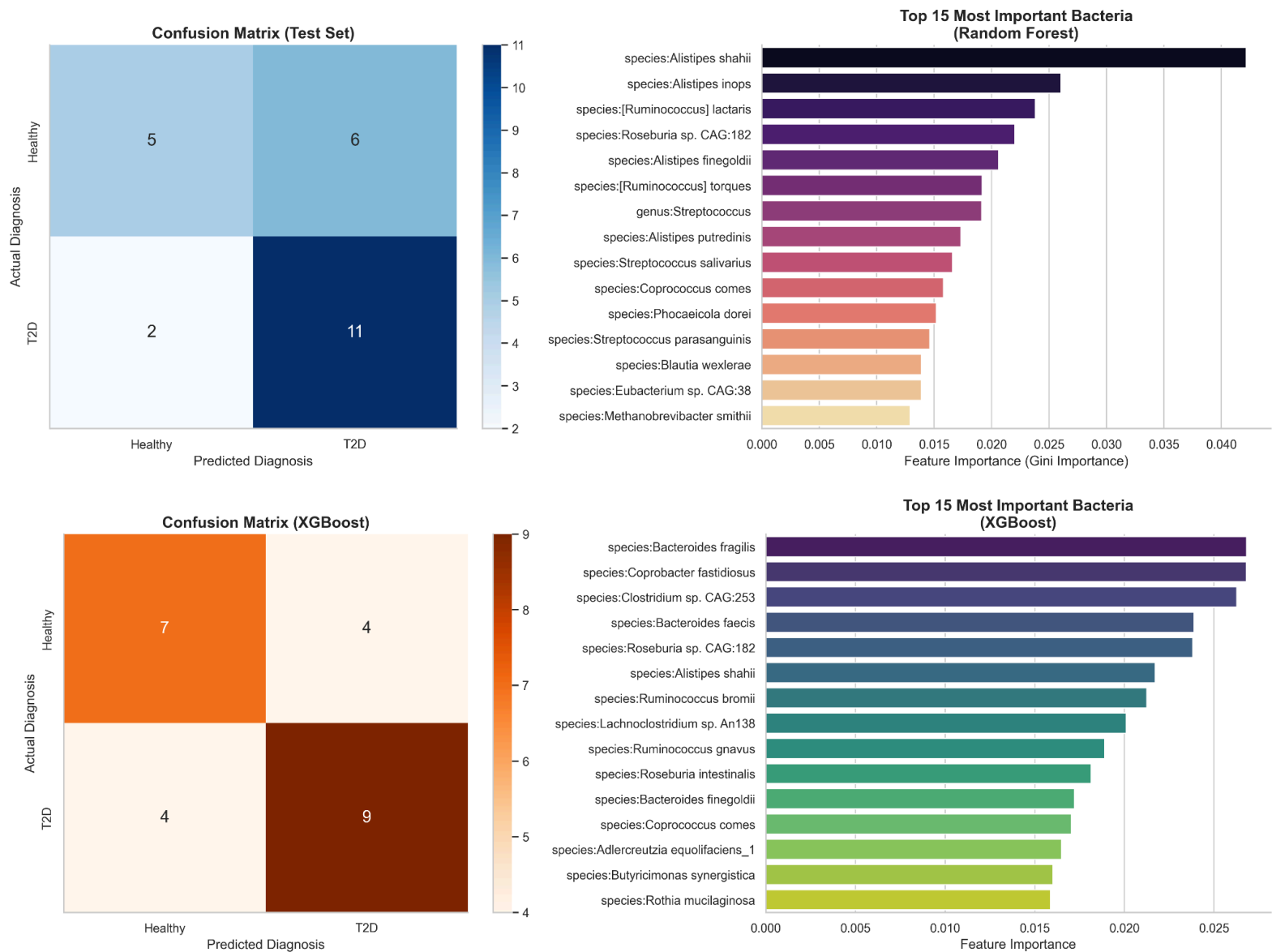
Predictive Modeling

We trained two tabular classifiers on the full processed feature set:

- Random Forest
- XGBoost

For the tabular models, the data were split into training and test sets using a stratified 80/20 split. Performance was evaluated on the held-out test set using ROC-AUC, accuracy, precision, recall, F1-score, classification reports, and confusion matrices.

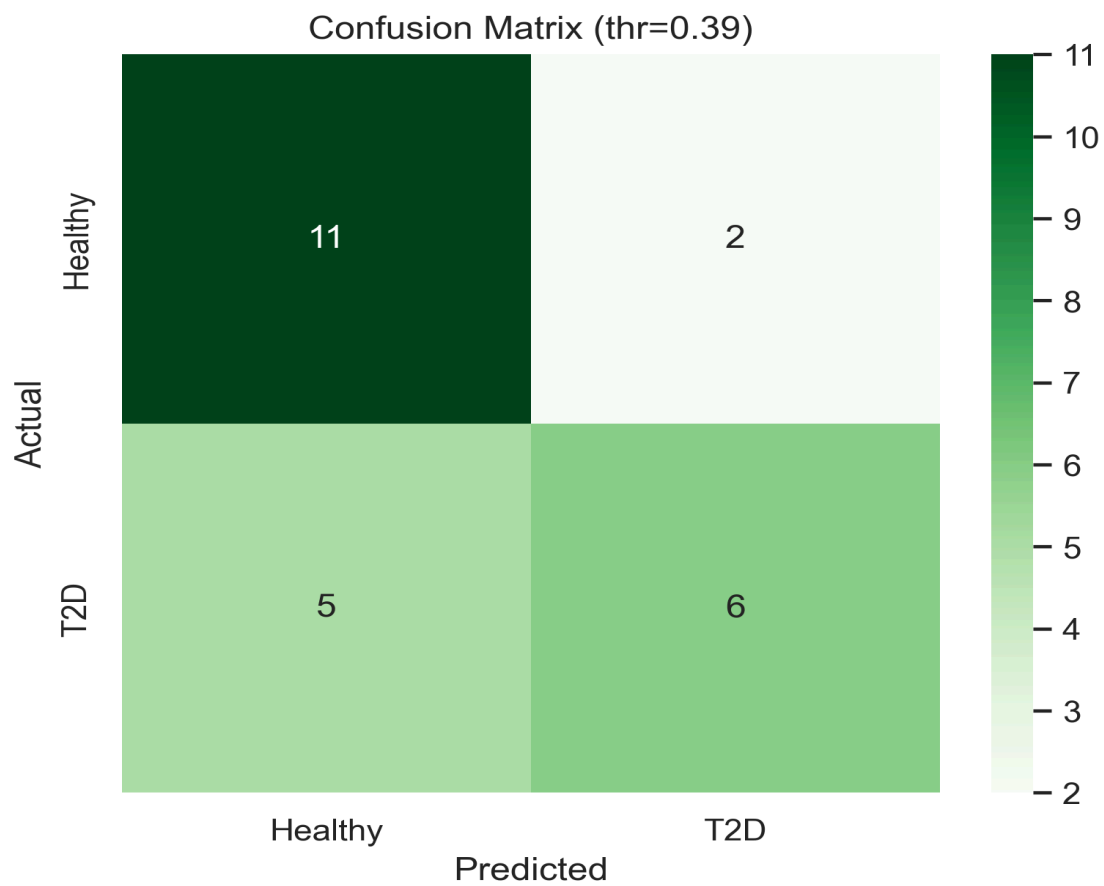
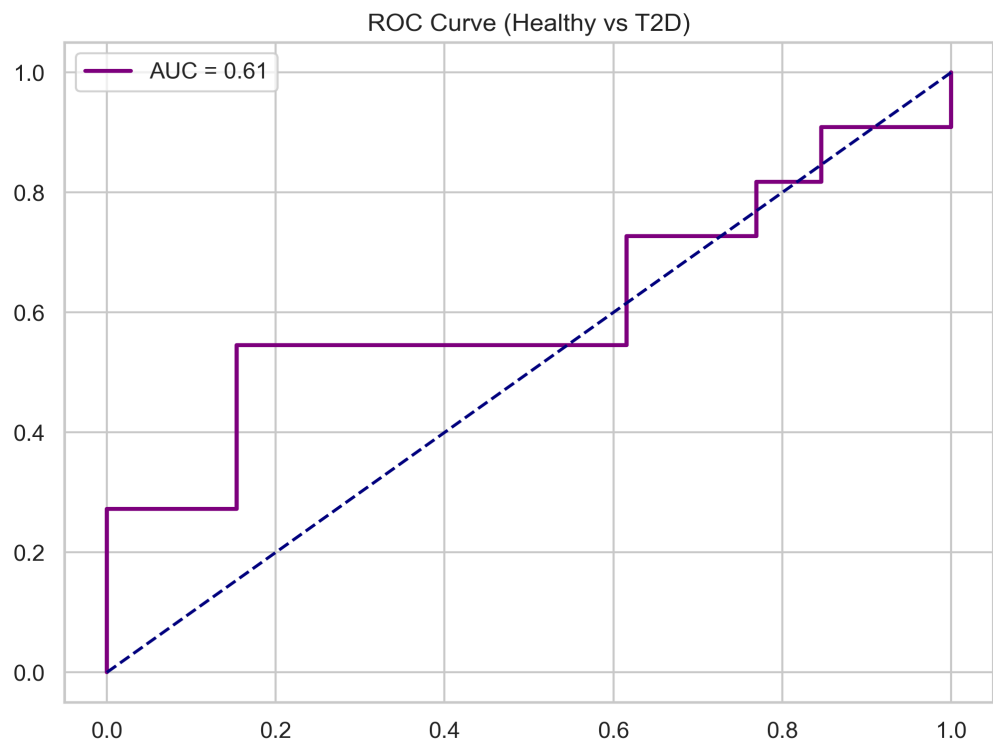
For interpretability, we extracted XGBoost feature-importance gain scores to identify taxa that contributed most strongly to the predictive model.



Taxonomy-Aware CNN

To evaluate whether a structured spatial representation could help, we transformed each microbiome profile into a compact image-like matrix with `micro2matrix` and trained a CNN using PyTorch Lightning. Optuna was used to tune

core hyperparameters such as learning rate, dropout, and batch size. The CNN was evaluated using ROC-AUC and threshold-dependent classification metrics, including a threshold selected by Youden's J statistic.



Post-hoc Reduced-Feature Experiment

Finally, we ran a follow-up experiment using a smaller feature set guided by the statistical analysis. Because the strict mi-Mic-based filter produced no directly usable features for this modeling step, the notebook relaxed the selection to the top 20 raw p-values and retained 19 taxa for retraining XGBoost and Random Forest.

Results

Exploratory Findings

The two cohorts shared a similar core of dominant taxa, especially *Faecalibacterium prausnitzii*, *Ruminococcus bromii*, and *[Eubacterium] rectale*. This overlap suggested that class separation was unlikely to be driven by a single overwhelmingly abundant bacterium. The dataset was also visibly sparse and zero-inflated, and the log transformation produced a more stable abundance range for analysis.

Differential Abundance

The Mann-Whitney plus FDR pipeline identified 16 significant taxa ($q < 0.05$). Taxa enriched in healthy controls included *Roseburia* sp. CAG:182, *Alistipes shahii*, *Alistipes finegoldii*, and *Bacteroides cellulosilyticus*. Taxa enriched in T2D included *genus:Streptococcus*, *Streptococcus mutans*, *Streptococcus vestibularis*, *Lactobacillus paragasseri*, *Blautia coccoides*, and *Enterocloster lavalensis*. The observed log2 fold-changes ranged from approximately -5.73 to $+7.55$, indicating strong but taxon-specific effects.

Taxonomy-Aware Validation

The global mi-Mic nested ANOVA was not significant ($p = 0.99998$). This result argues against a broad tree-wide microbiome shift separating T2D from healthy samples. Instead, it supports the interpretation that the disease-related signal is concentrated in specific taxa and in multivariate interactions among taxa.

Full-Feature Predictive Models

Full-Feature Predictive Models

Among the full-feature models, Random Forest gave the best overall performance:

- Random Forest: ROC-AUC 0.780 , accuracy 0.708 , precision 0.667 , recall 0.923 , F1-score 0.774

- XGBoost: ROC-AUC `0.720`, accuracy `0.667`, precision `0.692`, recall `0.692`, F1-score `0.692`

These results show that the microbiome carries predictive signal for T2D, although the performance remains moderate rather than near-clinical.

Model Interpretability

The top XGBoost importance rankings included `Bacteroides fragilis`, `Coprobacter fastidiosus`, `Clostridium sp. CAG:253`, `Bacteroides faecis`, `Roseburia sp. CAG:182`, and `Alistipes shahii`. Some of these taxa overlapped with the statistically significant taxa, while others appeared to contribute mainly as part of a multivariate predictive pattern. This supports the idea that the classifier uses a distributed microbial signature rather than a single biomarker.

CNN Results

The taxonomy-aware CNN underperformed the tree-based tabular models:

- CNN ROC-AUC: `0.608`
- At threshold `0.5`: poor positive-class behavior
- At Youden-optimized threshold `0.390`: accuracy `0.708`, precision `0.750`, recall `0.545`, F1-score `0.632`

Although threshold tuning improved the operating point, the CNN did not outperform the simpler tabular baselines.

Reduced-Feature Follow-up

Using the 20 selected taxa:

- XGBoost improved to ROC-AUC `0.741`
- Random Forest dropped to ROC-AUC `0.713`

This suggests that limited feature filtering may help XGBoost slightly, but aggressive reduction also removes information. The strongest result in the notebook remains the full-feature Random Forest model.

Discussion

The main conclusion of the project is that gut microbiome composition contains a real but moderate signal associated with Type 2 Diabetes. This signal is not consistent with a broad taxonomy-wide shift across the microbial tree. Instead, it appears to be feature-specific and multivariate, with several taxa showing differential abundance and with the best predictive performance coming from models that combine many taxa at once.

From a modeling perspective, the results favor classical tree-based methods over the more complex CNN representation. In this dataset, Random Forest was the strongest overall baseline, outperforming both XGBoost and the taxonomy-aware CNN. This is an important practical result: a biologically motivated deep-learning representation is not automatically better, especially when the cohort is relatively small and the feature space is sparse.

Biologically, the findings suggest that T2D is associated with enrichment of several taxa such as `Streptococcus`-related organisms and depletion of taxa including `Roseburia` and `Alistipes` species. However, these results should be interpreted as association rather than causation. The project does not prove that these microbes cause T2D; it shows that they are statistically and predictively linked to the disease state in this cohort.

The project also has several limitations. The sample size is modest, there is no external validation cohort, and the results depend on a single dataset and preprocessing pipeline. The CNN workflow is also likely sensitive to representation choices and dataset size. For future work, the analysis could be extended with repeated cross-validation, external validation on an independent cohort, integration of clinical covariates, and deeper investigation of confounders and microbiome interpretability.